Valuation: Part I

Relative Pricing

Relative Pricing

- Determine the price of an asset by comparing the market value of similar or comparable assets

- Important steps
  - Determine the comparable assets
  - Convert the market price of comparable assets into a ratio relative to the fundamentals
  - Acquire the pricing ratio by comparison
    - » Adjust all possible deviations

The Importance of Relative Pricing

- Relative pricing gains its popularity in the Wall Street

– Almost 85% of the stock research reports are based on the multiple of comparable stocks

– More than 50% of M&A pricing is based on multiples

– Multiples are usually the basis for making final investment decisions

- Why relative pricing

– Pricing is often a bargain.

Demands for Pricing

- On February 3, 2012, the director of the relevant departments of the CSRC claimed that the CSRC would study the supervision of the underwriting process, and will direct it to **rationally** set IPO prices. The CSRC is working on a measure to promote the fair pricing of new shares, which is to entrust a third party to publish the average P/E ratio of listed companies in various industries for investors' reference.
- IPO pricing is generally no more than 25% of the industry's P / E ratio.
- Since 2014, with a P/E ratio of 22.98 times, regulators almost ensured that every investor subscribing to new shares is profitable. The 22.98 or 23 PE ceiling was cancelled in year 2023

张劲帆，李丹丹，杜涣程，2020，IPO限价发行与新股二级市场价格泡沫——论股票市场“弹簧效应”, 《金融研究》, 第1期, 190-206。



新浪财经 会议讲座 > 中国经济50人论坛2019年年会 > 正文

方星海：新股上市首日限价的弹簧效应

	Flexible PE (2009.07.01-2014.06.30) Mean	Fixed PE (2014.07.01-2018.06.30) Mean	Difference
Maximum post IPO PE	63.9	94.6	30.7***
	(56.8)	(49.0)	(13.73)
Maximum post IPO PE minus	10.0	30.4	20.4***
Industry Average PE	(9.8)	(17.3)	(10.03)
Days for Post-IPO Stock Price	1.7	11.9	10.3***
Rise	(33.3)	(73.0)	(60.13)

注：括号内为 t 统计量，***、**、*分别表示为 1%、5%和 10%的显著性水平。

Unit：%	Equal Weighted			Market Value Weighted	
	Flexible PE	Fixed PE	Difference	市场定价发行	窗口指导发行
1 year abnormal return	-8.2***	-18.7***	-10.5***	-12.3***	-27.3***
t-value	(-7.33)	(-11.12)	(-5.17)	(-11.36)	(-20.75)
Number of Obs.	934	846		934	846
2 year abnormal return	7.5***	-12.8***	-20.3***	-5.0***	-27.6***
t-value	(3.43)	(-4.50)	(-5.66)	(-2.74)	(-13.22)
Number of Obs.	934	404		934	404
3 year abnormal return	29.4***	-13.8***	-43.1***	7.0**	-36.0***
t-value	(8.42)	(-3.98)	(-8.78)	(2.52)	(-12.86)
Number of Obs.	934	266		934	266

注：Abnormal return is to use the stock return minus the market return during the same period of time.

Understanding Multiples

- The definition of multiples
  - How to figure out the multiples
- Analyzing multiples
  - Know the distribution of multiples
- The factors of multiples
  - What determines the multiples?
- Using multiples
  - Determining comparable collections and adjusting deviations is not easy.

The Calculation of Multiples

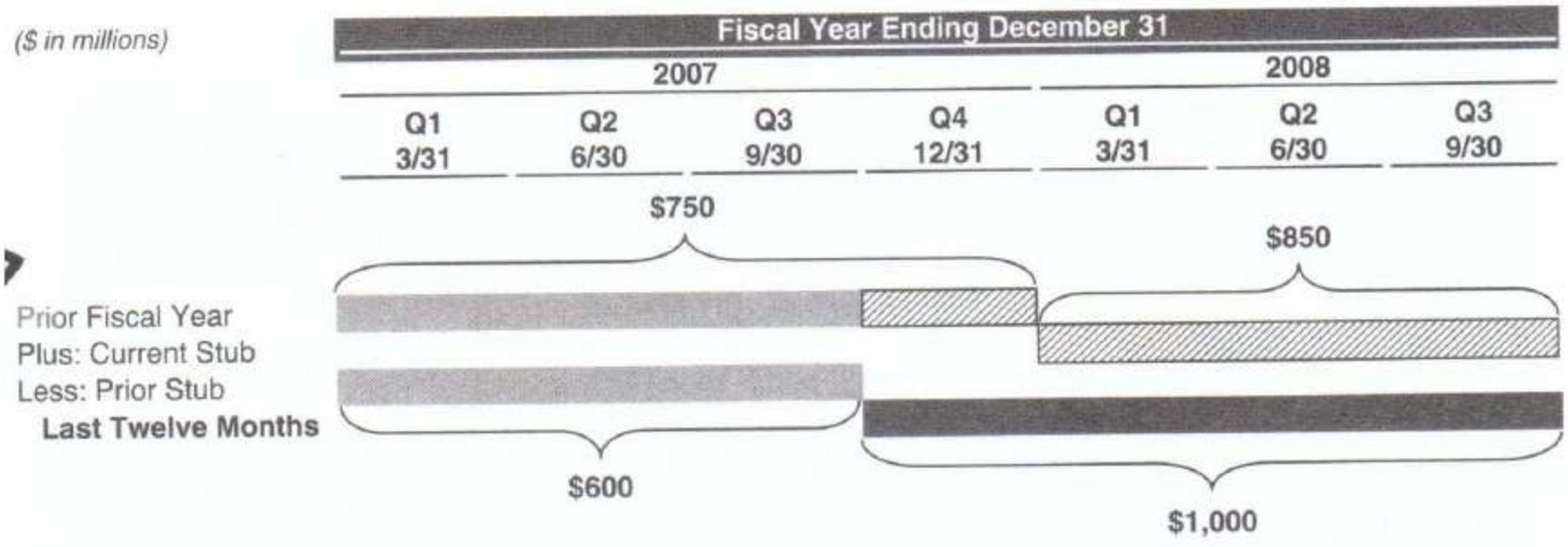
- **Some important financial data**
  - Earnings before Interest, Taxes, Depreciation and Amortization (EBITDA)
    - » =EBIT + Depreciation and amortization (DA)
    - » EBITDA Margin=EBITDA/Sales Revenue
  - Net Income
    - » =EBIT – Interest– Taxes – Non-controlling Interest(NCI)
      - – Preferred Dividend
    - » It is customary to calculate net income as earning per share(EPS)
    - » P/E ratio = Share price/ EPS
    - » ROE=Net Income/Net Asset
    - » ROA=Net Income/Total Asset

The Calculation of Multiples

- Updating recent financial data.
  - Data for the past 12 months (LTM, TTM)
    - » = Data for last financial year+ Data for the current year – Corresponding data for last financial year

Update Recent Financial Data

EXHIBIT 1.25 Calculation of LTM 9/30/08 Sales



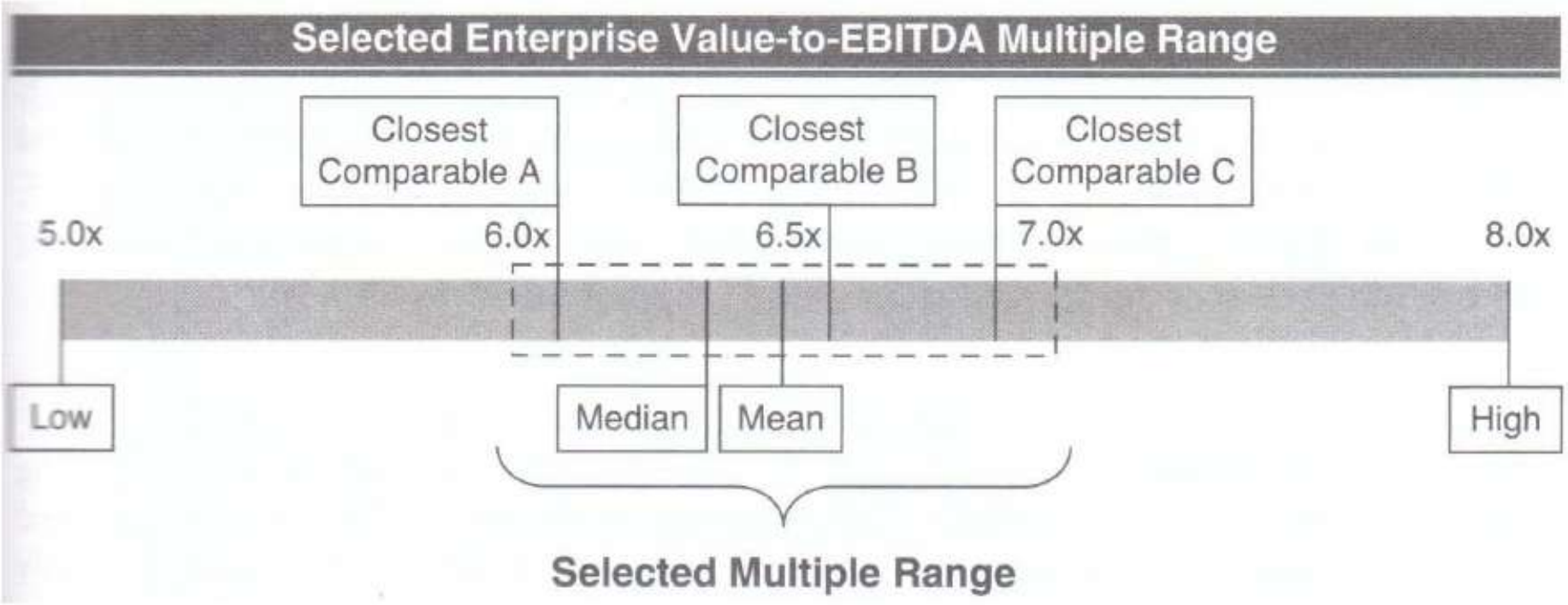
The calculation of multiple

- Multiples
  - P/E ratio
    - » =share price/earnings per share
  - P/B ratio
    - » =share price/Book value per share
  - enterprise value and EBITDA
    - » =EV/EBITDA
  - enterprise value and EBIT
    - » =EV/EBIT
  - enterprise value and sales
    - » =EV/Sales

Benchmarking

- Choose a comparable company
  - – calculate key financial ratios and multiples
  - – calculate the range of financial ratios and multiples of comparable company
    - » average, median, highest and lowest
    - » investigate and understand differences
- Benchmarking
  - – Establish the range of valuation
    - » in general, to find the mean and median of the most relevant industries

Benchmarking



Benchmarking

($ in millions, except per share data)

EBITDA	Financial Metric	Multiple Range	Implied Enterprise Value	Less: Net Debt	Implied Equity Value	Fully Diluted Shares	Implied Share Price
LTM	$200	6.50x – 7.50x	$1,300 – $1,500	(500)	$800 – $1,000	100	$8.00 – $10.00
2008E	215	6.00x – 7.00x	1,290 – 1,505	(500)	790 – 1,005	100	$7.90 – $10.05
2009E	230	5.50x – 6.50x	1,265 – 1,495	(500)	765 – 995	100	$7.65 – $9.95

Benchmarking

EXHIBIT 1.36 Valuation Implied by P/E – Share Price

($ in millions, except per share data)

Net Income	Financial Metric	Multiple Range	Implied Equity Value	Fully Diluted Shares	Implied Share Price
LTM	$70	12.00x – 15.00x	$840 – $1,050	100	$8.40 – $10.50
2008E	75	11.00x – 14.00x	825 – 1,050	100	$8.25 – $10.50
2009E	80	10.00x – 13.00x	800 – 1,040	100	$8.00 – $10.40

Benchmarking

($ in millions)

Net Income	Financial Metric	Multiple Range	Implied Equity Value	Plus: Net Debt	Implied Enterprise Value
LTM	$70	12.00x – 15.00x	$840 – $1,050	500	$1,340 – $1,550
2008E	75	11.00x – 14.00x	825 – 1,050	500	1,325 – 1,550
2009E	80	10.00x – 13.00x	800 – 1,040	500	1,300 – 1,540

Summary

The advantages of relative pricing

– Utilization of matket information

» Considering market growth, risk estimates, and market sentiment

– Relativity

» Easy to be measured and compared with other companies

– Fast and convenient

» Easy to build a computing platform

– In time

» Market information: updated every day

Summary

- The disadvantages of relative pricing
  - Utilize market information
    - » Market sentiment can lead to pricing bias.



Summary

The disadvantages of relative pricing

- Lack of comparable companies
  - » Baidu (IPO Aug. 2005 $12/share, at the end of 2007 $40/share)

- The price could be very different from DCF.

- Corporate individual factors

The most recent valuation of high-tech companies in China and US

	Forward P/E	P/B	P/S
Tesla	127.7	15.7	11.6
Netflix	42.5	18.3	11.7
Xiaomi	41.0	5.9	3.2
Nvidia	35.2	51.6	30.1
Apple	32.5	55.1	9.4
Amazon	31.9	8.5	3.8
Meta	28.4	10.2	11.3
BYD	21.2	6.2	1.4
Alphabet	20.3	7.0	6.5
Tencent	19.2	4.7	6.7
Alibaba	13.6	2.3	2.3
NetEase	13.6	3.7	4.6
Baidu	11.1	1.0	1.8
JD.com	9.4	1.9	0.4

High dividend stocks

Domestic Stock Market

股票名称	股票代码	股息率（%）	市盈率（PE）	市净率（PB）	行业特征
中国神华	601088	8.5	6.8	1.2	煤炭龙头，现金流稳定
工商银行	601398	6.9	4.5	0.6	国有大行，分红常态化
大秦铁路	601006	6.7	7.2	0.9	重资产运输，低估值
长江电力	600900	5.8	18.3	2.5	水电垄断，高股息可持续性
中国石化	600028	5.6	8.1	0.8	周期性能源股，PB低于1

HK Stock Market

股票名称	股票代码	股息率（%）	市盈率（PE）	市净率（PB）	行业特征
汇丰控股	00005	7.2	8.0	0.9	国际银行业务，分红政策透明
中石化冠德	00934	9.1	5.5	1.1	石油仓储，高股息但波动较大
越秀房产信托	00405	8.6	6.8	0.7	REITs强制分红，PB低至0.7
电能实业	00006	6.5	12.4	1.3	公用事业防御属性，PB适中
中国移动	00941	5.9	10.7	1.0	电信垄断，PB贴近净资产

Precedent Transaction Cases Analysis

- Pricing by using multiples of comparable companies in previous M&A cases to price.

- It is a challenge to find comparable companies
  - Sometimes different industries can be considered when there is similar
    - » Final product market
    - » Distribution channels
    - » Financial characteristics

Google bought Motorola Mobility LLC

- August 15, 2011
  - Google bought MOTOROLA with $12.5 billion
  - 63% premium
  - How did they figure out $12.5 billion?



Google bought Motorola Mobility LLC

- MOTOROLA has 24,500 patents.
- In October 2010, Novel sold 882 patents that are worth $450 million.
  – How much is each patent worth?
- If Google buys 24,500 patents for $12.5 billion.
  – How much is each patent worth?
- Conclusion
  – Google bought 24,500 patents plus a free mobile phone company.

How Are Patents Priced?

- Pricing of Kodak
  - Market value: On November 24, 2011, $310 million.
  - There are 1000 patents
  - If each patent can be sold for $510,204
    - » $510 million

How Are Patents Priced?

- **[Kodak's proposed $525 million patent sale plan was approved by a bankruptcy judge]**

  On January 11th, kodak proposed a price of $525 million to Intellectual.

  Proposal for Ventures and RPX to sell digital imaging patents was approved by bankruptcy judges, which made the company closer to coming out of bankruptcy protection.

Precedent Transaction Cases Analysis

- Steps
  - Select relevant M&A comparable companies
  - Collect relevant M&A and financial information
  - Calculate key data, ratios, and transaction multiples
  - Benchmark related mergers and acquisitions
  - Pricing

Precedent Transaction Cases Analysis

- Transaction details
  - Is the buyer a strategic investor or a financial investor?
    - » Strategic investors are generally willing to pay higher prices
  - What are the motivations of buyers and sellers?
    - » Is the seller in urgent need of cash?
  - Transaction procedures and nature
    - » The auction
    - » A hostile takeover
    - » Equal participation
  - Buying patterns
    - » Cash
    - » Stock

Calculate Multiples

Multiples

- P/E Ratio = Price per Share/EPS
- Market Value to LTM EBITDA Ratio
- Market Value to LTM EBIT Ratio
- Market Value to LTM Sales Ratio

Calculate Multiples

- Adjusted Synergy Multiples

- Examples：
  - Market Value = $1.2 Billion
  - EBITDA = $ 0.15
  - Market Value to EBITDA Ratio = 8

  - Assume that synergy effect will increase EBITDA for $30 Million
    - » Market Value to EBITDA Ratio = 1.2/0.18 = 6.67

Summary

- Advantages of Precedent Pricing
  - Leveraging Market Information
    - » Take Actual Acquisition Price into Consideration.
  - Timely
    - » Reflect market`s current M&A price, circumstances of capital market, and current financial situation.
  - Easiness
    - » Easy to compare with other transactions.
  - Objectiveness
    - » No need of hypothesis on future performance.

Summary

- Disadvantages of Precedent Pricing
  - Leveraging Market Information
    - » Market Sentiment may cause option pricing deviation.
  - Time Delay
    - » May not be applicable to current situation.
  - Lack of Comparable Transactions
  - Missing Transaction Details